CONNECTOR EXAMPLE - SOPHOS FIREWALL TRANSCRIPT

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L00 - INTRODUCTION

L00A-PRES: Introduction to Sophos FW Connector (03m:50s)

00:00-01:00
Hi and welcome to the *Stellar Cyber Open XDR Firewall Connector* module. I'm David Shirley from the Stellar Cyber Enablement and Training teams. The lessons in this module show how to configure, test, and validate a firewall connector. Using a Sophos firewall as the

example, we start with the configuration required in the firewall for the connector to communicate the IP addresses to be blocked by the firewall. Next, we will configure and test the connector in the Stellar Cyber Data Processor. We finished the module with four different tests using both manual and automated actions that use the connector to block IP addresses on the firewall. Throughout this module, there may be new concepts and terms that you may not be familiar with. That's okay and to be expected. The important thing is you become familiar with the terms and concepts described. Through the Stellar Cyber Open XDR training curriculum, your understanding and application of cybersecurity principles will deepen with each repetition, iteration, and example.

01:00-02:00
As you advance, you're on your way to becoming a cybersecurity superstar. Thank you for your dedication.

The *Stellar Cyber Open XDR Connector Sophos Firewall* module is made up of the following lessons:

- *Overview*. In this lesson we'll take a high-level look at where connectors fit into the Stellar Cyber platform from a network-architecture perspective. We will look at the list of different types of connectors as well as their capabilities and the extensive list of external actions available.
- *Connector Configuration*. This lesson covers the required configuration of the Sophos firewall, including the creation of IP host groups and firewall rules, as well as the configuration of the API and log settings. Following that are the steps to create a connector in the Stellar Cyber UI.

02:00-03:00

- *Connector Testing*. Now that the configuration of the firewall and the creation of the connector are complete, this short lesson outlines the testing of the connector to confirm successful communication.
- *Connector Verification - Respond Action*. This lesson is the first of the three lessons that verify the connector by using the as an administrator or a SOC analyst would in day-to-day operations, starting with two tests using the respond action's manual method to add IP addresses to be blocked in the firewall.
- *Connector Verification - Interflow Detail Action*. Following on from the first two verification tests of the connector using respond action, the verification test in this lesson uses the connector via the external action feature of an Interflow record, making the process simpler for the SOC analyst. The IP address to be blocked is pulled from the Interflow record.

03:00-03:50

- *Connector Verification - ATH Action*. The final lesson in this module is also the final verification test for the firewall connector. The previous three examples require administrator or analyst intervention, where this method automates the process, utilizing the ATH or automated threat hunting playbook capability of Stellar Cyber, the blocking or unblocking of IP addresses in the firewall is all automated based on matching certain criteria. This is a great example of the "R" in XDR extended detection and response.

Now let's get started. Happy learning!

L01 - CONNECTOR OVERVIEW

L01A-PRES: Overview (11m:25s)

00:00-01:00
The first lesson in the *Stellar Cyber Open XDR Connectors - Sophos Firewall* module is *Overview. Overview*. In this lesson, we'll take a high-level look at where connectors fit into the Stellar Cyber platform from a network architecture perspective. We will look at the list of different types of connectors as well as their capabilities and the extensive list of external actions available.

In this Overview lesson, we are talking about connecting our Data Processor or creating a connector in our Data Processor, to communicate with a firewall and do that via an API configuration. That is what we use our connectors for. From a big picture perspective, we've got our Data Processor or our platform up in that top right-hand corner. In the left top we have a Modular Sensor with a switch, some Windows servers, but what we're here to talk about today are these connectors.

01:00-02:00
We have the concept of either cloud connectors for connections directly to our Data Processor from technologies that are in the cloud. We can see there those cloud security services and apps whether they're identity or email or EDR. Or local technologies that we can create connectors for whether they are firewalls or Active Directory or vulnerability scanners. What is a connector? Stellar Cyber can parse log data. We know that Stellar can bring log data normalized and enriched into the Data Platform, but from an API perspective or a connected perspective, we're also able to support using the API connections to receive data from local SaaS and cloud-based security network and application technologies. That's just the receiving.

02:00-03:00

We're also able to push changes such as blocking a firewall or disabling a user. API connectors are developed by Stellar Cyber per request and are released with new versions of Stellar Cyber. We're about to go through and see a huge table of the already existing many, many connectors that are available to all of our customers today. The first thing we're going to look at are the types of connectors. On the right-hand side, that is just a screenshot from our Knowledge Base. We can see that our connectors in this table are grouped. Of course, that is just a screenshot of the very top of the table, but what we're going to see is a couple of pages here of the different types of connectors starting with cloud security. For example, if you had Prisma Cloud we could create a connector to work with that technology. Then going down the list databases with MySQL or Email with Proofpoint, Endpoint security with all our different EDR providers,

03:00-04:00

BlackBerry Cylance is an example there, the different firewalls so FortiGate or Palo Alto. Our example today will be a Sophos firewall, but there are many different firewall connectors. Honeypot like Thinkst or Thinkst Canary, IDP so identity like Active Directory or Entra ID, PaaS or platform as a service or something like Amazon Web Services, CloudTrail. Moving on to our second page we can have a remote host. We can SSH to a remote host and provide a plethora of different examples where we could run a script on that remote host. Software as a service or something like Office 365 in the cloud what we can connect directly to. SASE by Cato Networks is another way we can connect security, switch, or vulnerability scanners. Tenable: We've got a great lesson on the Tenable scanner from a Modular Sensor feature perspective,

04:00-05:00

web security, something like Cloudflare, or a webhook using our Universal Webhook Responder, another tool that opens up many different applications for connecting to technologies. Alright. Connectors have different capabilities. The collect capability for a connector. Source data we know we can pull information or we receive information from our log sources, but from an API perspective, we have the collect feature where our API or the connector can ingest data into Stellar Cyber. Then we have a respond capability, the ability to perform an external action. In the next couple of pages we'll see what all of those external actions are. Some connectors do collect. Some connectors do respond. Some connectors can do both. Depending on what that technology is, the best place to look it up is in the Knowledge Base to see what's available.

05:00-06:00

When you create the connector, if it's not available to respond or if it's not available to collect, that checkbox will be grayed out. Alert integration: Many of our partners that we have connectors for, they trigger their own Alerts in their technologies. It's very interesting

for us to receive those Alerts, but we want to normalize them when we ingest them. Stellar Cyber is able to ingest, normalize, and enrich those third-party Alerts. Then we apply our machine learning and statistical analysis engines, a direct mapping, or a combination of two to create Stellar Cyber Alerts with deduplication to reduce the noise and Alert fatigue. What we don't want is to have an Alert from an EDR come in and then we also trigger on that information so we deduplicate in those environments.

06:00-07:00

The Runs On: This is a pretty important idea. We have the choice to run, well, not always a choice. We have the capability to run the connector either from the Data Processor or from the Modular Sensor. What we really mean by "runs on" is which Stellar Cyber piece of technology is communicating with that end device. In our example, we have a Sophos firewall. That Sophos firewall has the capability to either run on the DP or run on the Modular Sensor. That comes down to what does the network architecture look like? We'll spend a bit of time to show why in our network we've chosen to run it on the Modular Sensor. Interval: The interval is only available for connectors that collect information. This is saying, "How often do you want Stellar Cyber to go and pull information from that connector"? That is the interval timer there.

07:00-08:00

Then our actions. Now we can see there are four pages of actions here of what's possible with all of those different types of connectors. Now of course these are not available for our Sophos connector. We have one capability available and that is the block IP. We are able to send an IP address to the firewall to be included in a group that was preconfigured that is bound to a rule that is already in the sort list for the rule. That's how that piece works. The rest of them are not available on the Sophos firewall, but we'll see what they are available for. In this example, we can disable a user or enable a user in Active Directory or Entra ID. Confirm Compromised: Again, Entra ID will see that Entra ID is a very powerful connector with a lot of external actions available. Dismiss Risk is another one. Run a script on that SSH host. Now that we've got an SSH host as a connector, we can run a save script on that.

08:00-09:00

The graphic you're seeing is the bottom of an ATH rule or an automated threat hunting rule. That is the last example. We will configure one of these playbooks. Sorry, I've been calling them rules. We'll configure one of these playbooks, an ATH playbook, to see how that works with the connector. Contain Host: If you have endpoints EDR of some sort we can manage or isolate those endpoints from within the Stellar Cyber platform. The big picture here for all of these external actions is they're all capable of happening from within the platform. The SOC analyst is able to do this from within the platform without having to leave it and go to another device or another technology or another console or another web interface to do

something. From within Stellar Cyber we can do this. We'll see a couple of different ways during our tests of how we can do that. Simply blocking an IP address on a firewall there's a number of different ways to do that.

09:00-10:00

Hide Host: If we have CrowdStrike's Falcon technology on that host. Update a device if it's running the Forescout connector with a JSON file. Initiate a scan: We'll see here whether it's a Sonic firewall capture or a SentinelOne technology running on that host we can initiate a scan, kill, quarantine, or remediate a threat. That is if SentinelOne is running on that end host. Disconnect, restart, or shut down a machine: Again, if it's SonicWall capture we can do all those different types of external actions. Final ones–remediate an email: The example is if you're running Barracuda's forensic and incident response system, we could remediate an email from directly within the Stellar Cyber platform. A webhook: Using our universal webhook responder, a very powerful tool that opens up capabilities broad and wide of what we can do.

10:00-11:00

Back to the Run On. We can see here that we have the ability to run the connector on the sensor or run it on the Data Processor. What we mean by "run" is, "What is the Stellar Cyber technology that's going to communicate directly with that other technology"? Where are the API calls going to be run from. We can see here they can either be run from the sensor or the Data Processor. If you do want to run your connector from your sensor, of course you will have to have created that sensor previously. Otherwise, during your initial configuration of your connector it won't be available. Make sure you create the sensor first, and then the connector if you want to use a Modular Sensor. Of course it's not required if you're using the Data Processor to run your connector. If the device you're connecting to is on premises, then we recommend that you run on the local sensor, on the local Modular Sensor.

11:00-11:25

Of course, if it's a cloud service then we recommend you run it on the Data Processor. Do check with the details of the connector in the documentation. You may have a particular need for your network architecture to run it one way or the other. We will go into detail about our architecture, so you can understand some of the choices that we have made.

The lesson, *Overview,* is complete. Thank you.

L02 - CONNECTOR CONFIGURATION

L02A-PRES: Connector Configuration (17m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Connectors - Sophos Firewall* module is *Connector Configuration. Connector Configuration.* This lesson covers the required configuration of the Sophos firewall including the creation of IP host groups and firewall rules as well as the configuration of the API and log settings. Following that are the steps to create a connector in the Stellar Cyber UI.

This lesson is all about configuring the connector. Let's start and look at what does our lab look like from a configuration perspective and what our IP addressing and host so that we can just become familiar with what it is that we're working with and what it is that we're trying to install. Let's start bottom right. We have our Stellar Cyber Open XDR platform, our Data Processor we also call that. 192.168.1. You can see the switch on the right-hand side is that 192.168.1.0. It is connected to another firewall to get out to the internet. We've put a host out there, 8.8.8.8, because we use that address during our testing and we'll see it in these presentation slides. In the middle, we have a firewall, the Sophos firewall. We can see there on the left-hand side it is connected to the 10.20.1.0 network. With an address of .199. On the right-hand side it is connected to the 192.168.1.0 network. I don't know what that address is. Oh! There it is. 124 is written on our screen, unimportant to us that address actually. Then further on the left we have a Modular Sensor: 10.20.1.15. That's going to be important. We've got our test host that we're going to be blocking all the time: 10.20.1.8. It's just a Windows 10 host running some simple ping tests for us. Okay. Our DP: .175. Our Modular Sensor on the 10 network: .15. Our firewall with a .199 on the 10 network and a .125 on the 192 network and our Windows host with a .8. It's an important piece that Windows host default gateway 10.20.1.199. Now we can see that our Sophos is also sending logs. We can see Sophos logs to Modular Sensor on port 5520. We will show you a screen of why port 5520. That's the parser port for a Sophos firewall but we will show you that. Then down at the bottom we're going to create a Stellar Cyber connector for the Sophos firewall.

03:00-04:00
That's the big picture of what we're creating. Let's just start with the **Runs On** ID right now. We have a choice with a Sophos connector for a firewall whether we run it on the DP or we run it on the Modular Sensor. That interface on the firewall, the WAN interface as far as the Sophos firewall is concerned, doesn't have any management capabilities turned on. You know it's connected to, well, it probably looks like a DMZ network here in this example, but there is no way to communicate. We can't manage the firewall from the 192 network and we certainly can't connect our API to the 192 interface. That's the reason we have the Runs On idea of having a Modular Sensor in that left-hand network. I mean the Modular Sensor is doing a bunch of other things, but in this example, when we configure our Sophos connector on the DP, we'll see that we're going to create it to say run on the Modular Sensor.

04:00-05:00
The communication will work like this. You will sign into the user interface on the DP, the 192.168.1.175 and you'll issue a bunch of configurations. Those configurations' requests for the firewall will be sent

to the Modular Sensor and we're going to ask the Modular Sensor to communicate directly to that firewall. Of course the firewall port, the .199 on the 10 network, is open for management capability. That's the reason in this architecture that we're going to run the connector on the Modular Sensor even though in this picture it looks like that connector is well let's say it's configured on the Data Processor, but it's run on the Modular Sensor. Let's look at some of the overview. This is not a Sophos firewall course, but we of course need to understand some of the configuration on that.

05:00-06:00
Some key capabilities. Now if we look at what's available we can see for our Sophos firewall, you can connect Stellar Cyber to firewall so that you can quickly and easily implement firewall rules on those firewalls from within Stellar Cyber while you are threat hunting. You can configure firewall actions or manually trigger firewall actions, from the event display. Now if there's a couple of collect, no we cannot collect information using the API. Respond, yes. When we say that we can implement firewall rules, we'll look at very specifically how we interact with those rules to achieve the blocking of a host. Native Alerts, well our firewall is not triggering Alerts. There's no native Alerts coming. Just what we've been talking about, we can see for this connector, we can run it on the DP or the sensor. In our environment we're going to choose Sensor. We'll see that choice when we configure it. Alright.

06:00-07:00
To configure the Sophos firewall connector there's a couple of things that we have to have: One, we have to have the firewall. We create a source and destination host groups. They must exist before we create the connector. We've got to create the firewall rules. We actually don't have to create the firewall rules to create the connector, but we do need those source and destination host groups. Then we need to enable the API configuration on that firewall. Now when we say we don't have to do it, well if you didn't then it's a bit pointless because it's not going to work. The next thing we want to do is on the Stellar Cyber DP and we want to add the connector, test the connector, and then we'll verify that connector's operation by running a number of tests on it. First, let's start with the Sophos firewall. We need to create host groups on that firewall. We need to create firewall rules that bind those host groups to the rules.

07:00-08:00
We need to reorder the firewall rules so that they're in the correct order we expect them to be in. The final piece is to enable API access. That's what allows our connector to communicate with the firewall. Then the last piece, which is not required, but certainly helpful, is to configure login. We want to have logs forwarded from the Sophos firewall to the Stellar Cyber DP. This helps us understand what's going on with that. Particularly, we want to see the deny logs come through that prove that our API is working correctly. Okay. What do we need to do? Create two host groups: one for source and one for destination. On the Sophos firewall you'll navigate to **System | Hosts and Services**, select the IP host group, and then choose **Add**. We need to create a couple of them. One: Give it a name. Give it a description, if you like, and then save.

08:00-09:00
We can see we've already created a couple of those. We've called them Stellar source IP or SRC IP group and Stellar DST IP group. We've got our source and our destination IP host groups configured.

The next thing to do is to go to **Protect** and **Rules and policies** and add some firewall rules. We're going to create two firewall rules in this example. You've got to give it a name and we've got to give it a group. The group is like where is this traffic and an example here is it's traffic to WAN in the Sophos firewall. The action is to drop, so we want to block the traffic. It's going to drop any traffic that matches this firewall. We want to turn on log in. It's particularly important that we want to see rules that are matched trigger a log and have them sent to our Stellar Cyber platform. Choose the source, choose the destination service, and click **Save**. It looks exactly the same.

09:00-10:00
The only difference here is in the previous firewall rule, we mapped the source to our Stellar SRC IP group. In this firewall rule, we're mapping the IP group stellar-DSTIP-group to the rule. One was for source; one was for destination. That is the reason we've got two different ones here. Alright. There they are. There are our rules. We've turned them on. We've ordered them correctly. We've got our two rules sitting above. It's top-down processing here. As the traffic comes through from LAN to WAN, we'll see those rules. Do I map the first rule? Nope. Do I map the second rule? Nope. Then I'll hit the last rule, which is the any any any permit. We can see there make sure that you I mean what we've done here is filter the rules and we're only looking at LAN to WAN zones.

10:00-11:00
If you expand that out then we can see our actual rules, drag those around to make sure you've got them in the correct order. How to turn on API. We select **System** then **Backup & Firmware**, and then there is an **API** tab. What we are required to do there is (1) enable it because it's not enabled by default. (2) We're going to add the IP address of devices that it is happy to accept API requests from. In this example, our 10.20.1.15 is our Modular Sensor. This is pretty important to know this because when we choose the in the creation of the connector, the Run On piece we have to choose the run on Modular Sensor. That's the only device that's going to be allowed to communicate to it, the 10.20.1.15. When you've done that, click **Apply.**

11:00-12:00
Logging. We do want to turn logging on because we want to receive information about what's going on with this firewall. We select **Configure | System Services**. We add a Syslog server. You can see there we've added our, again our Modular Sensor that is a log forwarder. It's got that feature running 10.20.1.15. You can see the port number we've got there, the 5520, that is for our parser port for a Sophos firewall. The log settings so make sure that you've either turned some or all of them on or pick. We've picked you can see that column there for Stellar Cyber. For the one we've created we've chosen to send us everything. That's the configuration so far for the Sophos. Now the configuration required on our Data Processor. We sign in. We go to **System** and then **Connectors**.

12:00-13:00
After we're in our **Connector** table you can see on top right we've got a button to **Create** a new connector and that will open a pop-up window. After we've got to that pop-up window, there's a bunch of drop downs that we can choose to go find the connector we're trying to work with. Now you can see from the **Category** you can choose Firewall, and then from the **Type**, you can go and find the Sophos firewall, or you can actually just go straight to the type and scroll through it. But you'll be

scrolling through all those other different groups of connectors. The **Category** what that does is is it narrows it down. It filters what's available for us to choose from to what are all of the firewall connectors I have to choose from. Choose your drop down, pick your firewall, find Sophos firewall, and then select that. The next thing to do is to give it a name. We have to give our connector a name. This is not the name of the host. This is just a name that exists within Stellar Cyber.

13:00-14:00
When we look at the table, we'll see this name existing, and we will see this name when we look in Interflow records as well. Give it a name, and then choose a tenant. Depending on whether you're a SaaS platform or you're an on-prem platform–on prem's have a root tenant, SaaS deployments you have to choose the actual tenant–that this connector is going to be bound to. We're running in this example an on-prem deployment, so we're going to just choose the root tenant here. But we can see there are choices in that drop down. Well, actually it's really a drop up. We don't want to cover up the next button, but you could choose other tenants in this example. The **Run On**. We've talked about this quite a lot about where it's going to run. Clearly the choice for this network architecture that we've made we're going to run on the Modular Sensor. That is the IP address that we told the Sophos firewall to expect API calls on.

14:00-15:00
In the **Run On**, choose the drop down, and if you've already configured it, which we have, we'll see our Modular Sensor: sc-modular-sensor in that list. Select your Modular Sensor. When we've done that, check your configuration. Make sure it's everything you want it to be, and then select **Next**. The **User Name**. This is how the API is going to sign into the Sophos firewall. The example here we're using is admin. That's the username that we've been signing into the device. Give it the password to the password you use to connect to the device and the IP address. In our example, it would be the 10.20.1.199. That's the interface that we're going to communicate to the Sophos firewall. We're not going to communicate on the 192 side, on the WAN side, Nothing is open There is no management capabilities opened on the WAN side of that device.

15:00-16:00
IP address: 10.20.1.199 would go in there, and then the important pieces here. These are the source IP group and the destination IP group. You've really got to think about these groups from the perspective of the firewall. We'll see when we go create these, when we go to push these IP addresses to the firewall, that it's from the perspective of the firewall. We created these groups already. We created stellar-SRCIP-group and stellar-DSTIP-group and they're the two that we will populate these fields with. Enter the source, enter the destination, and then port. This is the management port for the device. By default, and we know that from a security perspective it's good to change defaults on our security devices, particularly a firewall.

16:00-17:05
The default for our Sophos firewall is 4444, and that's the one we're using. Enter that port number or whatever port number it is that you use to communicate with your Sophos firewall. The last thing to do is check your configuration, and then choose **Next**. Then we're done. One last thing, select **Submit**. This is going to push this configuration and update our Data Processor. Now our connector has been

pushed to the Data Processor. From a configuration perspective, you'll see it in the table and what we're looking for is that big green splash bar: New connector was added successfully. That goes away after four or five seconds. It's just a splash bar. Then you'll see the connector that your newly configured connector in that table. The next lesson will be to go and test that connector to make sure our configuration was successful.

The lesson, *Connector Configuration*, is complete. Thank you.

L02B-DEMO: Connector Configuration (17m:00s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall* module is *Demonstration of Connector Configuration. Demonstration of Connector Configuration.*

In this demonstration, I will show you an overview of the configuration of the Sophos firewall and the required elements needed for the configuration, including host IP groups, firewall rules, and the API settings. Then we turn our attention to the Stellar Cyber environment and look at the existing setup, and then onto the creation of the connector in the user interface. Let's start with signing into the Sophos firewall. Here we are in the Sophos user interface. Okay. There are a couple of things to look at first.

01:00-02:00
We're going to look at the interfaces to see how we've set this up. We go to **CONFIGURE** and **Network** over to **Interfaces**. We can see here our Port1 is on the 10.20.1.199 network with a 24-bit mask or 255.255.255.0 and Port2 is our WAN interface. We can see that that is also connected. It's on the 192.168.1 network with an address of 124. It's got that via DHCP. It's a pretty simple setup. Ten network on one side, on the LAN side, and on the WAN side is the 192 network for this firewall. The building blocks for what we're about to do comes here under **SYSTEM | Hosts and services** and then **IP host group**. We've configured two of these IP host groups.

02:00-03:00
We've got one called stellar-dsp-group and the other is stellar-srcip-group. They are just names; there is nothing special about the names. There's no key syntax that it needs to be. It's just the name. But we do need to know that name, because we will use that during the configuration in the Stellar Cyber user interface. If I go into these, you can see, it's just a name with a description. I could give it a description. It's unimportant to me on this side of things. That is our DSTIP group and then the other one is just the Stellar SRCIP group. They're just alphabetically listed there. They are important to what we're going to use. We're going to see where they're bound in our firewall rules. If we go up to **Rules and policies**, and remember we're using the LAN and the WAN, if we go to **Source zone** LAN, **Destination zone** WAN, alright.

03:00-04:00
Here we are: Traffic to WAN is the headline there. I've previously created these two firewall rules and then ordered them above rule number five. We can see rule number five is our any any any permit.

Any source, any destination, any protocol, any port, from any source, any destination, any service is ID number five, or rule number five. I've created these two rules above that. In the ordering rules we start top to bottom and we can see the first one is the Stellar Cyber source SRC. This is using the source of LAN with the group stellar-srcip-group. The other one below it is with a destination to stellar-dstip-group.

04:00-05:00
Let's just look at the source version of that. We can see here that's just the name of the rule. It is a drop action. We are going to log this. That's important. We're going to log this traffic so it will be sent via our Modular Sensor as a port forwarder onto our Data Processor. We'll see the traffic from this firewall being sent. Particularly for this rule we can see we've got logs sent. The source is the LAN zone. The source network and devices is this group: Stellar source group. Then our destination is the WAN zone to anywhere for any servers. Just remember that firewall for the source. The important piece here is that that group is part of the source, if it matches that it will match this rule.

05:00-6:00
If we look at the destination, slightly different, so it's got a different name of course **Action** is **Drop**. We are logging firewall traffic for this firewall rule. The source is the LAN zone, any source network or device going to the WAN zone that must match stellar-dstip-group. Okay. There are two IP groups we've got configured. We can see that we've bound those two IP groups to some rules that are already in our firewall list. We can see here they're both drop rules. If you look at that really carefully, I can hardly read it. I know that one's log. I don't know what that one says. What does it say? Something. I can't even read it. I think they're all grayed out, but log is the important one we're interested in to know that they will be logged. Of course the **Action** is **Drop**. Now let me just jump over to our documentation.

06:00-07:00
If we go to our docs and we search for parser we probably want the top result. There it is: Log Parser Ports. I'm going to do a browser search and search for Sophos, and I will find a Sophos firewall here. Let me close that. Okay. This is the one we're looking at: Sophos firewall. That's the one. The port we're interested in is 5520. Just remember that as we go back to our Sophos firewall interface. We're going to go down to **System services** under **CONFIGURE**. We're going to go to the **Log settings** tab, and we're going to see what's previously being configured. We have a Syslog server over here. We've called it Stellar Cyber 10.20.1.15. That is the address of our Modular Sensor, which of course is a log forwarder, and we're sending to port 5520.

07:00-08:00
That is the port that Stellar Cyber's expecting to receive these logs on and so we can parse, normalize, and put these logs into an Interflow record and forward it to the Data Processor. If we just look at that we can just say, that's our address. That's the port we're sending to. These are just a couple more configurations regarding what we're actually sending–**Severity level** or **Facility**. They're just different pieces of the logging information in Sophos. The final piece we're interested in is the API. This is the part that's the communication piece on the Sophos side to allow communication from the

connector when we configure it on the Stellar Cyber DP side. To do that, we come down to **Backup & firmware**, which feels like a funny place for it to be.

08:00-09:00
Then API, and we can see here API is configured. It is enabled, the allowed address, so it needs to have a source address that matches, is 10.20.1.15. That is our Modular Sensor. When we create this connector, we'll get a choice that we can run this connector on the DP itself or on a Modular Sensor. We're going to run it on a Modular Sensor. We'll show you why when we look at a diagram of the layout of this network. Okay. Now back to the DP. Let's log in. There are just a couple of things to look at. First, let's look at our sensors so we can see the sensor we're interested in. We are interested in this sc-modular-sensor. There is our IP address: the 10.20.1.15.

09:00-10:00
That is the one that is approved by Sophos to communicate with the API. Our Log Forwarder Service is green; we can see over here. The log traffic that is arriving from Sophos will come to our Modular Sensor and then off to our DP. We can actually just go down here to our **Log Forwarder Summary** and we can see what is coming in. 5520. There it is down the bottom. That is the Sophos traffic that's coming in. We can already see there's logs that are coming in. Okay. What else to look at? **System**, we want to look at our Data Lake. We can see the address we're talking about here. Our address is 192.168.1.175. That's how we manage this device. It's just as an address I want you to be aware of because we're going to use this as a destination to ping from one of our hosts during this test.

10:00-11:00
Next on the list is for us to create the connector. We go to **System | Connectors** under **Integrations**. We can see we've got a list of connectors already here. Some of them are happy, some of them are unhappy, and a **Create** button. This is what we're after. Create the category. There are a couple of different ways you can do this. You can either search the category and it'll be very easy for us to find firewall, and then amongst the firewalls, we can find Sophos. That would be one way to do it. The other way to do it would be to click **Create** and just go straight to the type, but you'd have to scroll through all of the different connectors available until we get to firewall, and then all the bottom of the firewall. There's a couple of different ways you can get there. Now this connector is a respond only. It's not a collect. You can see you can't even turn that on.

11:00-12:00
We're collecting the information from the Sophos firewall via its Syslog server. We don't need that. The **Name**. This is just the name that we give it that's going to show up when we're going to **Investigate | Threat Hunting**. We can see this when we look at our tables and our pie charts. This is the name that's going to show. Let's call it "SophosFirewall." That sounds good enough to me. Okay. The **Tenant**. We can either choose to run it in the root tenant or we can choose to run it in one of our other tenants. We're going to choose the root tenant in this example. Now **Run On**. We got the choice of the Data Processor or a Modular Sensor. One moment, let me just pull up a diagram of our network to explain why we would do this a particular way. Alright. Here's a look at the lab network that we're going to use for this.

12:00-13:00
We can see our DP of 192.168.1.175. That's on the right-hand side of our Sophos firewall. On the left-hand side of the Sophos firewall is the 10.20.1 network, and we can see we've got a Modular Sensor there: 10.20.1.15. The Sophos firewall itself, 10.20.1.199 is on the left. 192.168.1.24 is actually an irrelevant address for us. The Windows host on the ten network: the 10.20.1.8. That Windows ten host has a default gateway of 10.20.1.199, which is the Sophos' LAN side. Any traffic from that host, 10.20.1.8, must go out through the Sophos firewall to arrive in the 192 network or out to the internet. That's the setup there. That's why this is important.

13:00-14:00
Here is the reason we want to know the difference between the **Run On** the DP or the **Run On** the Modular Sensor. Our data platform is going to send commands to the Sophos Firewall to block some IP addresses. That's the overall goal of this connector. The Sophos firewall has no way to communicate to it on the WAN side. All of that's turned off of course. You wouldn't want that turned on. Any admin communication must happen on the LAN side, on that 10.20.1 network side. The way the communication will work is the DP will issue the command, we'll create these block rules. We'll see what those look like. It will send it to the Modular Sensor and the Modular Sensor will communicate that to the firewall. That is just one reason to do this. The other reason for having the Modular Sensor there on the left of course is we've got some Windows Server hosts.

14:00-15:00
We've got some Linux Server hosts there, and they both run-in server sensors. All of those will communicate via that Modular Sensor which has the Aggregator feature turned on. Not important for what we're trying to do here with the connector, but we do want to know that we want to run this connector on the Modular Sensor. The communication between the Sophos firewall and the Stellar environment will be via that 10.20.1 network. You may well be browsing to the 192.168.1.175, the Stellar Cyber Data Processor, but the communication to the Sophos firewall will be via the Modular Sensor. All the comms will go from the DP to the Modular Sensor and then to the firewall. That's why we want that configured that way. When we come back to our configuration here, we want to run it on the Modular Sensor. That's the one we looked at before. Select **Next**.

15:00-16:00
Now **User Name**. These are important. This is how we sign into the Sophos firewall. Admin, the password, the IP address. This is the IP address of the Sophos firewall that we're going to communicate to its API on and that is the 10.20.1.199. Now these source groups we saw those before. These are the groups that Stellar needs to know what they are. Remember we called these stellar-srcip-group. We call the destination one stellar-dstip-group. Then the final one is port. The port number we need is from this host, the Sophos firewall.

16:00-17:00
To do that, we just go down to **Administration** and **Admin and user settings**, and this is the one we're after: the **Admin console HTTPS port**. That's the port we're going to use to communicate. At the moment, it's 4444. Back to here and it's just 4444. That is actually the default one. Of course, you could change it. If you did choose to change it being that you don't like to use defaults to be a bit

more secure, you would need to match whatever's there in this connector. Then select **Next**. You've completed the configuration. The only thing left to do now is click **Submit**. You can see the first thing we see is the green box: New connector was added successfully. What I want you to look for here is that there is no green dot. We'll come back and test this in the next lesson.

The lesson, Demonstration of Connector configuration, is complete. Thank you.

L03 - CONNECTOR TESTING

L03A-PRES: Connector Testing (04m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall* module is *Connector Testing. Connector Testing*. Now that the configuration of the firewall and the creation of the connector are complete, this short lesson outlines the testing of the connector to confirm successful communication.

We've made all the required configuration changes to the Sophos firewall. One: To create those IP host groups, create the firewall rules, turn on logging, and turn on our API, and add to our API our Modular Sensor to accept API calls. Then on the Stellar Cyber platform, we've created that connector. We've created it with the IP address, username, and password of that Sophos firewall. We've added the host groups, the source and destination host groups, that's stellar-srcip-group and stellar-dstip-group.

01:00-02:00
We've added both of those to the connector. We've saved the connector. The last thing we saw was the big splash. Now that we've done that, when we edit or change or add a connector, we can run a test to validate that the connectivity is there. Now you can see in that note, the test validates only the authentication and the connectivity, it does not validate the data flow. This will validate that we've got the IP address right, that we've got the username right, that we have the password right. It's not necessarily going to validate that we've got the IP groups right, those host groups correct. Let's go ahead and see what this testing looks like. It's a very short lesson before we get onto running the actual validation tests.

02:00-03:00
Navigate to **System** and **Connectors** to bring up the table of our connectors. There is our connector in the top line. Locate the connector that you've added or you've modified that you want to test. Then on the right-hand side I mean look, first of all the **Status** button is green. That's a good sign. On the right-hand side we can see that there's a **Test** button. Select **Test** to initiate the test of the API, the connection. The test runs immediately. As soon as you click that button, the test will start running, but you can only run one test at a time. If you have multiple connectors in there, only one connecting test can be done at a time. Stellar Cyber conducts the basic connectivity test for the connector. You can see there it's now gone from being a **Test** button to an **In Progress** button. The big splash screen we like reads, “Test Success!”

03:00-04:05
A successful test indicates that we've entered all the connector information correctly, meaning the IP address, the username, and the password for this connector. Now other connectors have different requirements. If it's an Active Directory connector, it's certainly going to test more than the username and the password. There are a lot more features that I could test it in that one. In this one I believe it's just testing access to the API using the IP address, the username, and the password. We talked about the green dot before. A green dot is a good sign that it's healthy, and that our connector is ready to go. You can see in there **Collect**: not available. **Respond**: Yes, please. That's how we're going to send our IP addresses and insert them in those host groups that are bound to those firewall rules that will block our host, our IP addresses.

The lesson, *Connector Testing*, is complete. Thank you.

L03B-DEMO: Connector Testing (04m:10s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall* module is *Demonstration of Connector Testing. Demonstration of Connector Testing.* Now that the configuration of the firewall and the creation of the connector are complete, this short demonstration outlines the testing of the connector to confirm successful communication.

In this demonstration, I will show you the process to test the connector starting with the Stellar Cyber user interface and then checking the logs in the Sophos firewall. Pay attention; this is a really short lesson. Let's start with the Stellar Cyber user interface. Okay. We've just created our firewall connector, our Sophos firewall connector. Navigate to **System | Connectors**. We can come in here and see the **Status** currently is not available.

01:00-02:00
If we go to Sophos firewall, and just have a check we can see our settings in here. It's going to run on the sc-modular-sensor. The device is 10.20.1.199. That is the address of the LAN side of the Sophos firewall. Then our IP groups: We've got stellar-srcip-group and stellar-dstip-group for the destination IP groups. They all look good. The first thing we want to do is just go back and run the test. Select the **Test** button. After we've found our line in the connector table, you can see it's in progress. We can only run one of these tests at a time. This is not testing the actual pushing of firewall rules or host into those rules; this is just testing the connection.

02:00-03:00
We can see there we've had a test success with the pop-up window, and we've now got a green dot under **Status**, which says healthy. So far, so good from our side. If we just go back into the Sophos firewall one more time we can see **Status**, **Last Activity**, and **Last Data Received**. Well, we haven't done anything yet. We'll check those fields later when we actually push some records. Let's jump into the Sophos firewall and look to see the result of that test we just did. Into our firewall. We haven't set

our storage key. We're just going to ignore that for the moment. Okay. Straight from this home screen up the top right here we have **Log viewer**. We're going to **Log viewer**.

03:00-04:10
The **Log viewer** we want to look at–this is currently set to **Firewall**. We want to choose **Admin**. What we can see here is the last admin activity was the GUI was at the admin logging in from 10.20.1.13. That's the device I'm currently on. You can see here, but just below that so how long ago maybe a minute or so, not even, maybe 50 something seconds before, was an API call. It's in the log group for API. It was successful. The username was admin. The source IP 10.20.1.15. That is our Modular Sensor's IP address. We can see from both sides–from the Stellar side that it is working and we can see from a log that's been created on the Sophos firewall–that that API is working. So far, so good.

The lesson, *Demonstration of Connector Testing*, is complete. Thank you.

L04 - CONNECTOR VERIFICATION RESPOND ACTION

L04A-PRES: Connector Verification - Respond Action (15m:15s)

00:00-01:00
The next lesson in the S*tellar Cyber Open XDR Connector - Sophos Firewall* module is *Connector Verification - Respond Action. Connector Verification - Respond Action.* This lesson is the first of the three lessons that verify the connector by using the connector as an administrator or a SOC analyst would in day-to-day operations, starting with two tests using the respond action's manual method to add IP addresses to be blocked in the firewall.

Alright. Just the beginning of our four tests we can see up in the top left-hand corner there, we are test number one. This is the very beginning of our tests that let's get a network architecture layout of what we're going through here. Our Data Processor, our DP 192.168.1.175, that's what we're going to browse to. It's what we use to configure our connector.

01:00-02:00
It's what we're going to use to go through these tests to initiate the blocking of an IP address using the connector and pushing it to the Sophos firewall. Why don't we just take a look? Let's walk through the flow. The connector is configured. We're going to be in the UI on .175. We're going to issue a block command. We'll see the different ways of doing that. The Data Processor, the DP at 175, will push that request to the Modular Sensor 10.20.1.15, and that's what will communicate directly to the Sophos firewall on .199. As we saw, it's the Sophos firewall's configuration that said I'm only going to listen to API requests, from 10.20.1.15 our Modular Sensor, and we've then got our Windows host 10.20.1.8. That's the one that's going to be doing the pinging. We've got a host out there on the internet at 8.8.8.8.

02:00-03:00
It just happens to be a DNS server for Google that will respond. We've got two responses that we're going to ping. We're going to ping the DP and we're going to ping that 8.8.8.8 address. In this first test you can see on the left-hand side we're going to block the IP host 10.20.1.8. We should see those pings

fail and the method we're going to use respond action, so one of the manual methods to do that. Let's walk through how to do this. Just an overview of the Sophos firewall, what's been configured. We can see that the firewall has been preconfigured with IP groups and firewall rules using those IP groups. We've created our stellar-srcip-group and our stellar-dstip-group. Those IP groups have been bound to these two rules. Those rules are sitting higher than the any any any configuration.

03:00-04:00

In this configuration, we're looking at from the LAN zone to the WAN zone. That's the direction of these firewall rules. There are our bound IP host groups. You can see in those rules. That's what we're going to see get triggered when we issue these commands from within Stellar. The first thing to do is navigate to **Respond | Actions**. This is where we'll do the first piece. We're going to use a firewall action as you can see here. **Respond** and then **Actions** and then **Firewall Actions**. The next thing to do is to create, so we're going to create a manual firewall action. Okay. From that pop-up window after pressing our **Create** button, we can see that there's a number of things that we need to configure. The first one is we've got to choose the firewall, which we're doing this firewall action for. So in the drop down you select your firewall. In this example, it's the Sophos FW that we selected.

04:00-05:00

We're going to choose what sort of action it is from the drop down. Is it an add or is it a delete? The **IP Address**: This is the IP address that we're going to block. The example is our IP host, our Windows host 10.20.1.8, and the direction. I want you to really think about this. This is from the perspective of the firewall. The firewall is sitting there and saying I'm going to block incoming. You could think of that as, that's the source address we're going to block for. We're going to block 10.20.1.8 coming into the firewall from, well we know that that device is on the LAN side going to the WAN side, and that's how it'll match those records. Then we choose a time. The time can be in minutes, in hours, it can also be forever. We've got a choice of that. Then the last thing to do is click **Submit**.

05:00-06:00

As soon as we do that we've created the configuration and that begins the process of that information being pushed to our firewall via our connector, that API interface on this Sophos firewall to block that IP address. We've done the configuration. We get a nice, big splash banner–it's a temporary splash banner–to say that the configuration's been done. Then there's the **Status**. If you keep clicking the refresh button to refresh the contents of the table, you'll see that that **Status** will go from **Waiting**, to **In Progress**, and then to **Succeeded**. That's the steps that that will go through. What we'll see from the host's perspective is that the pings were happily working. You can see there that 10.20.1.8, our Windows 10 host, was pinging 8.8.8.8 successfully, and 192.168.1.175 successfully,

06:00-07:00

but as a result of us doing that action, that firewall action, that host has been blocked and those pings are now failing. We can see it's going to block both of them in this example because we blocked the source, the host from doing anything. Not just the pings but if that host was doing anything else from the LAN zone to the WAN zone, it would also be blocked. Okay. After five minutes, remember we chose five minutes in the timer, after five minutes that block will be released and communications will resume. Now 5520. We said we'd have a closer look at this. This is the

documentation. This is the parser port table from the Knowledge Base documentation. What we can see here is that for a Sophos firewall, it is configured to be bound to port 5520,

07:00-08:00

which means when the Sophos firewall sends its logs to our Modular Sensor with a destination port of 5520, that Stellar Cyber knows exactly what to do with that log file. It knows exactly how to parse that file. It knows what the field names are. It knows how they're formatted. It knows everything about the structure of that log file to be able to then normalize that file, send it to the Data Processor, and have it enriched as an Interflow record in our Data Lake. You can see there 5520. That's an important number. First, remember we'll see that come when we look at these Interflow records. Then the other piece to look at here is well where is this going to go? You can see there that it says that firewall log data will go to the **Traffic** index. To find log data from the firewall we will look in the **Traffic** index. Looking from our UI to confirm what's going on here. We're going to validate the firewall has denied traffic from 10.20.1.8.

08:00-09:00

To do that, we will go into **Investigate | Threat Hunting**. Let's navigate there. Change the index to **Traffic** as we just saw in the documentation, all of the log files from the Sophos firewall being received on port 5520 are sent to the **Traffic** index. To view that log, let's put in a search. We can see the search term we're putting in there: msg_origin.source of fw_sophos. If we go back to our documentation, you actually see there is that the field name fw_sophos, and it is part of the group firewall. When we come back to here and see fw_sophos as a message origin source, that's the reason for that. The capital A and D is to say we want to match all these criteria. We're going to match the FW Sophos and a source IP of 10.20.1.8.

09:00-10:00

and an action, a firewall action, of deny. That's what we're looking for in our table. We can see there after we've put in that search that all the records visible counted in our pie charts or in our bar chart or in the table below, all of those records are deny logs from the firewall. If we're to open up the Interflow dictionary by selecting **Open Interflow Dictionary** from that page, and then choosing the field name dstip, we can see that the balloons open up all of the different destinations that that firewall was going to that are now being denied by that firewall. These are all the logs coming in that show all the different destinations going. Of course the two biggest ones there are our 8.8.8.8 and our 192.168.1.175 for our example.

10:00-11:00

In our second example, our test number two, we're going to do a slightly different test. Same network architecture. We don't need to go through that again, but what we can see is the same pings. Right? Our 10.20.1.8 is being 8.8.8.8 and 192.168.1.175. The difference here is instead of blocking the source host 10.20.1.8, we're going to block the destination going to 192.168.1.175. That host, our Windows host should be able to get to 8.8.8.8, but not 192.168.1.175. The method we're going to use here is a respond action. First of all, this verification of the configuration will use the manual process of blocking an IP address in the firewall using the firewall actions. We navigate to **Respond | Actions** to get to where we want to go to.

11:00-12:00
We want to go to the **Firewall Actions** tab, and then we want to create a new one. The difference here in our configuration, we're still going to choose the same firewall. We're still going to add. The difference here is it's not the source address of 10.20.1.8 It's the destination address of 192.168.1.175. The other difference before we had this as incoming, now it's outgoing. Remember this is from the perspective of the firewall. This is an outgoing, leaving. It's a destination IP address is how we've got to think about this. After we've done that choose how long. In this example we can see we've chosen 10 minutes, not five minutes. After we've finished all that, we click **Submit**, which will then push that configuration into our table. We can now see that temporary splash screen saying that the configuration's been submitted. We have to start waiting to go through the **Status**.

12:00-13:00
The waiting, in progress, and then succeed. It is what we're looking for. What will we see that happens? Well, our 10.20.1.8 pinging the 192.168.1.175 will fail, but we can see that that's failed because that's the destination that's been blocked. We didn't block the source, so our 10.20.1.8 going to our 8.8.8.8 continues to work, so no problems there. It'll only block one of them: the destination of the .175, not the destination of the .8. After ten minutes, like we had configured in the previous one for five minutes, this one we just configured for a duration of 10 minutes. After 10 minutes, that IP address that's been put into the IP host that's bound to the firewall rule will be removed and communication to 192.168.1.175 will be allowed again.

13:00-14:00
You, as a SOC analyst or an administrator that's looking to block these, you have to choose a time period of like, first of all, why are you blocking it? You need to do some investigations. Something's happened. Something's being triggered. There's been a breach. You decide how long do I need to block this for to do some more investigation before I can unblock it. Remember the choice was you can do it in minutes, hours, days, or forever, or do permanently. You'll have to manually undo it. Alright. Let's go look at the Sophos firewall. If we go into that firewall, and we go into the IP host, we can see that an address has been added to that group. Because an address has been added to that stellar-dstip-group, it is the group that's bound to the destination firewall that's used in firewall rule number seven. That's how we've ended up having our host been blocked to the destination of 1.175. We can see there any LAN source, the destination, if it matches in that group, which now it does because we've the API or the connector has added that IP address into the group, for any service, including our ping, it will be denied. To validate that from our user interface on Stellar, let's go to **Investigate | Threat Hunting**, change the index to **Traffic**. This time we're going to do a slightly different search criteria. Same as before: message origin source of fw_sophos. We're only looking at traffic that's originating from our Sophos firewall, but this time we're choosing the source host of 10.20.1.8. We're not choosing the deny. We want to see everything in there. What we can see is both the allow traffic and the deny traffic from those logs from the firewall in that table.

The lesson, *Connector Verification Respond Action*, is complete. Thank you.

L04B-DEMO: Connector Verification - Respond Action (21m:05s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall* module is *Demonstration of Connector Verification - Respond Action. Demonstration of Connector Verification - Respond Action.* This lesson demonstrates verifying the connector by using the connector as an administrator or a SOC analyst would in day-to-day operations. Starting with two tests using the respond action's manual method to add IP addresses to be blocked in the firewall.

In this demonstration, I will show you two different tests using the manual **Respond | Actions** method. This method allows the administrator or SOC analyst the ability to block any IP address, in any direction for both a limited amount of time or indefinitely. We will use a Windows host and the Ping utility as a simple way to show this functionality.

01:00-02:00

Let's start with a look at the network diagram showing us the topology, devices, and hosts we are going to use. Here's our first test. We're going to ping from our Windows host. Our Windows host down there: 10.20.1.8. We're going to ping 8.8.8.8. That's a host out on the internet. It's actually Google's DNS address. We know that will respond. That host 10.20.1.8 is also going to ping our DP on 192.168.1.175. Both of those pings of course have to go through the Sophos firewall. The default gateway of that Windows host is the LAN side of the Sophos firewall. Any communication, whether it's to the 192 network or the 8 network out on the internet, must go through that Sophos firewall. That's what we're going to do.

02:00-03:00

As it says there the method, we're going to use the **Respond | Action**. It's a manual method of blocking IP addresses directly from the platform. That's the key here that we're not asking any administrator. I'm not on the phone. I'm a SOC analyst and I'm not jumping on the phone trying to find a network configuration person to ask them to put a firewall rule in to block an IP address because there's malicious or bad things happening from a particular address. I can do all of that from within the platform. Let's go look at the next piece of this picture. Here is our Windows host. We can see on the left we're going to ping 8.8.8.8 and we're going to do it continuously. On the right-hand side we're going to ping 192.168.1.175 and we're going to do that continuously as well. That's our Windows host with the default gateway. Actually, we could probably just show you that quickly. Let's do it on the right-hand side. If we do an `ipconfig`, we can see our default gateway there of 10.20.1.199.

03:00-04:00

That is the Sophos firewall. All communications from this host must go through that Sophos firewall to get anywhere. We're going to `ping 192.168.1.175 -t` for it to be continuous. Let's kick these off. There is no reason why they can't be going. You can see communication is happy, so through the default gateway to the 192 network, through the default gateway to the next default gateway, the next hop and off to the internet of 8.8.8.8. Let's just check our firewall rules. If we come back to our Sophos firewall, we're just going to check our rules. Navigate to **Rules and policies**.

04:00-05:00
The firewall rules as we know the **Source zone** is going to be LAN and the **Destination zone** is going to be WAN. If we expand those out, these are the rules we're looking at. The pings are currently working, right? They're matching this. It's from the LAN side. That's correct. Is there anything in this Stellar source IP group? Well, currently there's not. If I go down to **Hosts and services**, select the **IP host group** tab and look at our source group, there's nothing in here. There's nothing that's matching in there. As far as those rules are concerned, we're not matching. Let me just open those up again. Okay. Great. Let's keep them. Nothing matching here. Sorry, nothing matching on rule number six, going to anywhere. Rule number seven, LAN going to a destination. Well, we could go check-in that destination IP group as well.

05:00-06:00
I should have done that while I was here before. We'd go from the destination, and we can see nothing in this configuration. Nothing is matching those first two rules. All the traffic is getting to rule number five, which is any any any any any and it's showing as "Accept." That traffic's parsing. That's why our traffic here is currently succeeding. If we just go to that docs one more time, we remember that the Sophos firewall is coming in on 5520. It is going to be listed as a fw_sophos. It's going to be of type of firewall. What we're interested in here is data goes to the indicated index based on the log type. If it's a firewall, then it goes to traffic. That's what we're expecting to see in here. Let's just go into our DP and check that we're seeing some logs.

06:00-07:00
Well, we saw that when we looked at the sensor we could see that they were coming in, but let's actually jump in to **Investigate | Threat Hunting**. We go to the **Traffic** index. Let's unclick **Alerts**, go to the **Traffic** index. We're looking at the last 15 minutes. Here is our Sophos firewall. We can see there's already something coming in here. If we add to that, we can see some traffic coming in. There's a couple of deny's there from earlier. NETBIOS-4s. Okay, I'm not sure what those are from, from different network devices. Here is our device 10.20.1.8 going out to the internet somewhere using port 443 and allowed. Off it goes, no problem there. Alright. The first test we're going to do, let's just make sure we have no filters.

07:00-08:00
Clear all so that we're looking at all of our traffic. The first thing we're going to do is block 10.20.1.8 as a source going through the firewall. When we look at our host, both of these pings should fail because we're going to block the host as a source. Let's go and do that. As we said, we're going to go to **Respond** and **Actions**. This will show us the last hundred actions. We're going to go to **Firewall Actions** and create one. Let's select the **Create** button. The firewall, while there's only one, if there were multiples you could choose from this drop down. The **Action**: We're adding a rule, we're adding something to our groups. The **Address**: This is our host. I'll type it in: 10.20.1.8. This might be a little confusing at first.

08:00-09:00
The way you think about the direction. We've got a choice of incoming and outgoing. I want you to think of that from the perspective of the firewall. This is traffic coming into the firewall, and it's

coming into the firewall from 10.20.1.8. We're going to block it for a duration of five minutes. Select **Submit**. We've got the action was successful. We can see down here first of all the **Status** is "waiting." If I keep hitting this refresh button over here, let's see it go. It's going from "waiting" to "in progress." I imagine there's now communication by that API to the Sophos firewall. I refresh. We're waiting for one more status update. There's our final update: "succeeded." You can see that there is now a **Revert** button.

09:00-10:00
That would be used if I wanted to undo what I've just done. Let's jump in and have a look at what else we can see here. First of all, let's go back to our host. We can see now both of those pings are failing whether we're trying to get on the left-hand side out to the internet or on the right-hand side just simply through the Sophos firewall to the WAN side, the 192 side. They are both failing. Back to our user interface. Okay. Now that's failing. Let's see if we can find some of that traffic. Let's go **Investigate | Threat Hunting**. We know it defaults to the **Alerts** index. We've got no filters here. We're going to change from the **Alerts** index to the **Traffic** index. We can see all of this. I'm going to type in a search. We could type it in or maybe we could do it down here. We could say, "I want the source to be from the firewall."

10:00-11:00
Let's filter for that it's got to be from the Sophos firewall. We want the source to be our 10 network. Actually, we've got it down here in the table. I could right-click on this–the 10.20.1.8–and add as an including filter. What we've got is it must be Sophos firewall and from that source. I want to add one more. I want to say and the **Action** is deny. Okay. We've now got a couple of those. Let's go look at what we can see here. If I go look at one of these, look this is the one we know about, right? Here's our ICMP going from 10.20.1.8 to our 192.168.1.175. There is the other one directed to 8.8.8.8.

11:00-12:00
They're the two that are failing. If we go back to our lab, still failing. Okay. Let's have a look at one of those. We've got **Action**. There's all the Denies. Let's look at one of these. Let's look at the 192. They are the key fields. Let's have a look at the **Details** fields. We've got an **Action** of deny. That's what we expect. Now remember, we're getting this record has come as a log from the firewall, from the Sophos firewall. Destination IP: Well, that's what we're expecting. It's trying to get to 192.168.1.175. Let's look for the engine name. We can see that the record has arrived via the Modular Sensor because it's a log forwarder for the firewall. Here is our firewall policy ID. Is it ID rule number six that's triggered this record?

12:00-13:00
Let's go look. Rule number six: It's this one that's triggered that record. Right? Okay. Let's keep going. The message origin source: Let's have a look for that. There it is. The message origin source is the firewall Sophos. That's what we're expecting to see there. Protocol one with the IP protocol, ICMP is Protocol one. Let's find the source IP: 10.20.1.8 and the Stellar client address: 10.20.1.199. That's where this record has come from. That is our Sophos firewall. That's the IP address that we're seeing there. The final one to look at is the Stellar ingestion port 5520.

13:00-14:00
Let's just check if our host has returned. It's still timed out. One more thing we could look at back on the firewall, we could actually go to **Hosts and services** and look at IP groups and look at Stellar source. We can see here our 10.20.1.8. Now that's what's been added. When we did our firewall action, that's what it did. That pushed our address into that group and that group is bound as we know to that firewall rule number six. Our firewall rule number six has got that Stellar source group. That's why traffic has been dropped and we've received a log for it. Alright. It looks like five minutes have gone by and our traffic is working again.

14:00-15:00
If I jump back into our UI, if we go back to **Respond | Actions | Firewall Actions**, we can see that the **Revert** button is gone. It's expired now. That five minutes, when we did our create, and we only chose for five minutes, there are all the things that happened in that five minutes of real time. Okay. Let's set up for test number two. Alright. For test number two, we're going to do the same sort of test, but slightly different. The difference here will be we'll still do those same ping. Our Windows host 10.20.1.8 will ping both 8.8.8.8 out to the internet and 192.168.1.175 happens to be our Data Processor, but something on the 192 network and something out in the internet on the eight network.

15:00-16:00
The difference with this test is that instead of blocking the source, we're going to block the destination. We're going to block the destination of 192.168.1.175. What we should see is that the 8.8.8.8 should continue to work, but our 192.168.1.175 should fail. Let's start with that. To start there, we're going to go into **Respond | Actions**. Actually, let's just check that our hosts are happy. Both of them pinging away, no problems there. To **Respond | Actions** we're going to go to **Firewall Actions** and then select the **Create** button. There's still only one firewall. It is still an add. The address is 192.168.1.175. That's the one we're going to block.

16:00-17:00
This time it's going to be **Outgoing** because this is from the firewall's perspective that we're not going to let this–we're not going to let any communication go to that firewall and we'll do the same for five minutes and select **Submit**. Now we're going to wait. It says here green at the top, but this is the top rule here we're looking at. We've got "waiting." We'll hit the refresh button over here to see it cycle through to "in progress." Now we can see it has succeeded. Our **Revert** button comes up there. If we want to undo what we've just done. Let's go check our Windows host. We have been blocked. The reply, the 192.168.1.175 is not working, but 8.8.8.8 is still happily working. That's what we expected.

17:00-18:00
Let's just check our **Investigate | Threat Hunting** and see what we can see over there. **Investigate | Threat Hunting**. We know to change to the **Traffic** index. Alright. Now let's filter for our firewall, our Sophos firewall, and let's put a filter up here. Actually, we can go down here. Here is our destination. Actually, we don't want to do destination. We want to do source. We want to filter for our source. We're going to add our source as an including filter. Now what can we see down here? We can see some denies because being denied going to 192.168.1.175, but we're still seeing a bunch of allowed. There's a bunch of allowed traffic still going through.

18:00-19:00
Now if we look back at one of the 175. Let's go look at the detail of that and look for our firewall rule in there. We go to **Details**. Let's scroll down and see if we can see our firewall. Firewall policy ID seven. Okay. If we go look at that, in our firewall. Seven is this one, is the destination. Any host on the LAN side, if it matches going to the WAN side, and it's in this group, so let's go check that group. If we go to **Hosts and services**, we go to **IP host group** and this time we're looking at our destination. There we see our 192.168.1.175. Do we just jump back to our DP? Another interesting place to go look to see where are all the destinations that are going.

19:00-20:00
Currently remember we are looking at, if we look at our **Filters**, we are looking at only **Traffic** is the index, coming from the firewall, and coming from our 10.20.1.8. If we go into the Interflow dictionary, and we look at say, destination. I mean, look. Here's the first one. You can see all of these 526 records, 304 of them is the yellow, is the deny, and 133 is the allow. But if we were to say let's look at the destination IP dstip. This gives us a bit of a balloon. You can see the majority of our traffic is going to 192.168.1.175, but not right now. At the moment, we could see that traffic is timing out, but our 8.8.8.8 traffic is continuing.

20:00-21:05
Jumping into the Interflow dictionary is a great way to visualize all of the destination addresses from that 10.20.1.8 host. Just to confirm that our host is all happy now, we're getting pings from both sides. The 8.8.8.8 continued yet we're now seeing a successful connection to the 192.168.1.175. The last thing to do is just to check that our action is complete. Navigate to **Respond | Actions**. Select the **Firewall Actions** tab. We can see that it has expired. That is the end of test number two.

The lesson, *Demonstration of Connector Verification - Respond Action*, is complete. Thank you.

L05 - CONNECTOR VERIFICATION INTERFLOW DETAIL ACTION

L05A-PRES: Connector Verification - Interflow Detail Action (06m:40s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall* module is *Connector Verification - Interflow Detail Action. Connector Verification - Interflow Detail Action* following on from the first two verification tests of the connector using **Respond | Action**. The verification test in this lesson uses the connector via the external action feature of an Interflow record. Making the process simpler for the SOC analyst, the IP address to be blocked is pulled from the Interflow record.

In this test number three, to see our firewall connector, our Sophos firewall connector, same network topology layout, slightly different test. In this test, we're going to have our host 10.20.1.8 ping both destinations again 8.8.8.8 and 192.168.1.175.

01:00-02:00
The difference here is we're going to block the source, so we're going to block the entire source 10.20.1.8, to any destination. But the difference here is instead of using **Respond | Actions** we're going to use an Interflow record to do that. We'll go into the Interflow detail of a record and use the **Action** and **External Actions** from within there. Confirming that test three will use an Interflow Detail Action to block the host using the firewall rule on the Sophos firewall. The difference in the way that we're going to do this rather than going to **Respond | Actions** and manually creating a configuration to block, we're going to find a record that has our IP address in it. We're going to put a search in for a source IP of 10.20.1.8 and a protocol of one.

02:00-03:00
That will find our ICMP traffic. From the **Actions** column in the table we're going to select the "**i**" for more information on one of those records. It will open up this pop-up window. In this pop-up window we can see a couple of things. One: What we've already done is selected the **Actions** tab, which is further down in our pop-up detail of our Interflow record. We can see there that we also have a **Block IP** button listed under **External Actions**. The buttons that are listed under the **External Actions** are the ones associated to the record. For example, **Disable User**, **Block IP**, **Run a Script**, they all have users associated within this record. If the record didn't have a username associated to it, we would not see **Disable User** show up in this Interflow record. What we're going to do is an external action of **Block IP**.

03:00-04:00
In the pop up of the pop up, the first thing we need to do is select our connector. In this example it is the Sophos firewall, and then the next is the IP address. Now what's different here to the configurations we've done before with **Respond | Actions** is we used to have to type in the IP address that we were looking to block. We typed in either the source address of 10.20.1.8, or we did in one example I think the second test, we typed in the destination of 192.168.1.175. In this example when we configure this or action this from an Interflow record in the **Details**, in the **Actions** tab, and the **Block IP** button. What we're shown here when we click on the drop down for IP, is a list of IP addresses that are contained in the record. We can see here in our example the 10.20.1.8 and 192.168.1.175,

04:00-05:00
those addresses are both available in this record. If you're a SOC analyst looking to block something because you've detected a breach or you've seen a problem and you want to put this block in, you don't have to be familiar with the IP addressing of the network, you just need to know that the IP addresses that are contained in this record that's associated to the breach that you've done your threat hunting for, are the ones to block. You can choose either the source or the destination here or either address. Then our choice is **Direction**, the same **Direction** as the manual process before incoming from the perspective of the firewall or outgoing from the perspective of the firewall. Then

we're going to choose a **Duration**: how long are we doing this for? In this example, it is for five minutes. The last thing to do is select **Submit** to push that through.

05:00-06:00
Now, the moment we've done that it'll be like our **Respond | Actions** that we will get the pop up, the green temporary pop up to say the action was successful. What we can now also do is from within that **Actions** tab. We haven't left the detail of the pop-up window of our Interflow record. We're still on the **Actions** tab and when we select the **View Action History** button, we can actually see a record that that action was created for this Interflow record. We can see there that Sophos firewall block incoming on. The only thing it doesn't give us is a time, but because it has a time we will see that it will expire and that will also be populated in this record. If we want further confirmation we can go back to our **Respond | Actions**, not to create an action but just to view it in the table. Then finally, we can see the confirmation from our host that both IP addresses were blocked because we blocked the host,

06:00-06:40
the source, the host as a source, and not one of those addresses as a destination. Same as before because we set it for five minutes, after five minutes, that will expire and the block will be removed from the firewall or that IP address will be removed from the host group on the firewall allowing any traffic to continue.

The lesson, *Connector Verification Interflow Detail Action*, is complete. Thank you.

L05B-DEMO: Connector Verification - Interflow Detail Action (09m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall* module is *Demonstration of Connector Verification - Interflow Detail Action. Demonstration of Connector Verification - Interflow Detail Action.* Following on from the first two verification tests of the connector using **Respond | Action**, the verification test in this lesson uses the connector via the external action feature of an Interflow record. Making the process simpler for the SOC analyst, the IP address to be blocked is pulled from the Interflow record.

In this demonstration, I will show you another section of the UI more commonly utilized by a SOC analyst than an administrator–to block an IP on the firewall using the connector. Using the detail of an Interflow record, most commonly an Alert, to block an IP address contained in that Interflow record.

01:00-02:00
Let's start with a look at the network diagram to show us the topology devices and host we are going to use. Alright. Here for test number three, we're going to have our host, our 10.20.1.8, it is going to ping both of those destinations again, 8.8.8.8 and 192.168.1.175, both communication through the Sophos firewall. Then we're going to block the source IP, our 10.20.1.8, but we're going to use an

Interflow record to do that rather than that manual action that we were doing before. Let's jump into the UI and see what we can see in there. Alright. Log in. Alright. Here we are. Navigate to **Investigate | Threat Hunting**.

02:00-03:00
Change the index from **Alerts** to **Traffic**. We're going to type in a search. We're going to look for source IP, SRCIP of 10.20.1.8. Let's start with that. Great. We've got some records there and traffic. Let's do "AND" for ping. Capital A-N-D proto:1. ICMP. Great. Now we've got a couple going on there. Actually, let's go back. We need to turn on those pings in our host. Let's turn those on. That is our 8.8.8.8. That is our 192. Okay. They're both happy. Back to our UI, and we could probably do a refresh. If you just hit the little magnifying glass here, it will refresh that page. Alright. We've got some traffic coming in. We're looking at the last 15 minutes.

03:00-04:00
We're going to look for a record. We could use one of these records because we know that they have IP addresses in here, but more than likely the analyst will be in here looking for an Alert. Let's remove the protocol one from our search and just search for a source IP of that. We're going to change from our **Traffic** index to the **Alerts** index. So far we have zero Alerts associated to our source IP, but we're only looking at the last 15 minutes. Let's blow that out to 24 hours. Okay, great. We've got a couple of Alerts. We have External Firewall Denial Anomaly, Uncommon Application Anomaly. Let's just do the firewall, our External Firewall Denial Anomaly. Now what we do is hit the "**i**" select the "**i**" for more information on this. These are Interflow records as much as the other record was an Interflow record, this just happens to be an Alert Interflow record.

04:00-05:00
More commonly, this is where your SOC analyst will be. We hit the "**i**", and up comes our Alert detail. Now what we're looking for here is under **Actions**. Well, actually let's just **Key Fields**. In **Key Fields** we can see that we've got a source IP and a destination IP for this host. If we go to **Actions**, we can see one of the external actions is to block an IP. The difference here compared to what we were doing before when we were doing **Respond | Actions** is this: I don't have to remember the IP addresses. I can come in here and block IP. The connector is our Sophos firewall so that's the same as respond. Now the IP address, I don't type this in. It's already in here. We know that's our source address, 10.20.1.8, and that was the destination address. We're going to block our source. I don't have to choose it. It's bound to this record.

05:00-06:00
The **Direction** is coming into the firewall, and I'm going to do it for five minutes. Alright. When I select **Submit**, let's really quickly jump over. We'll just wait for this to run through. We should get our green banner any moment. Alright. There's our green banner showing that the action was successful. If we just close this out. Actually, let's open another tab. We'll go to **Respond | Actions**, and then select the **Firewall Actions** tab. Here we are. We're currently looking at this one. This is what was added from our record. Now if we go back to that record and open up I believe it was this one, we can go in here to **Actions** and look at the **View Action History**. We can see there Sophos firewall block incoming. That has actually happened.

06:00-07:00
Let's just go check our hosts. Yep. We are no longer successful. We've been blocked on the firewall. Let's go back to the UI and unblock that. If we come back here and select the **Revert** button so we know that that only happened. If I revert, I see a green banner stating that the Firewall action was reverted successfully. What it means by successful is that it's going to undo this. It's going to remove that IP address from that IP host group on the firewall. We can just do a refresh. Waiting is now in progress, and in progress is now succeeded. That's done. We should be able to go and have a look at our IPs, and they are happy again and pinging along. All successful. Instead of waiting for the five minutes we actually hit that **Revert** button to say well maybe I made a mistake and I shouldn't have done that. You go ahead and undo the block that's been put in place.

07:00-08:00
The last thing I just want to show you is let's have a look at the Sophos firewall. I just want to show you the logs in there of what we can see from the API configuration perspective, not configuration API, the use–the use of the API. We just have to wait for it to come in waiting for this banner to pop up to close. Skip for now. Okay. If I got the top here to log viewer, I think we had a look in here before. Let's look at **Detail**, but we don't look at all modules, we want to de-select **All** and only choose **Admin** and then select **OK**. What can we see in here? We can see these ones are all API. Actually the last one's GUI. That's probably us logging in right now. That's us logging right now. Is this the correct time? 4:07. This Windows host has a difficult time in keeping track.

08:00-09:00
That is the time. Four fifteen, this is us logging in. But what just happened? Let's look at the last couple of APIs. If we go back here We've done a couple Right? We've had an API successful. That was the first test we did where–I can even go back a little further: API 6:14, 6:12. Let's just look at this one. Let's highlight this one and just look at this one. You can see here it was an API log. The subtype type was admin. That's the group we're looking at. We're seeing here a user admin from the source 10.20.1.15, which is our Modular Sensor, use the IP host group Stellar source IP group. We can see that was–we added or this API call added a record to Stellar Source Group.

09:00-09:35
We can see that that was done. We can't see what the record was. We don't see our 10.20.1.8, but we do see that it was added to that source group, so we know that was us. There you go, a bit more verification of that API call from the connector in Stellar.

The lesson, *Demonstration of Connector Verification - Interflow Detail Action*, is complete. Thank you.

L06 - CONNECTOR VERIFICATION ATH ACTION

L06A-PRES: Connector Verification - ATH Action (10m:50s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall*, is *Connector Verification - ATH Action. Connector Verification ATH Action.* The final lesson in this module is also the final verification test for the firewall connector. The previous three examples require administrator or analyst intervention, where this method automates the process, utilizing the ATH (or automated threat hunting) playbook capability of Stellar Cyber. The blocking or unblocking of IP addresses in the firewall is all automated based on matching certain criteria. This is a great example of the "R" in XDR, that is, extended detection and response.

In our last of our four tests we're going to use the same network topology, so there is no need to go through all that.

01:00-02:00
The difference will be that the first three tests required an administrator or a SOC analyst to perform a function. What will be different about this test is we're going to use an ATH playbook, or automated threat hunting playbook. Based on traffic conditions or based on something that is triggered, we can then automate that blocking of a firewall. Let's look at how we can do that. The same tests here: There is a slight difference actually. The test number four starts off with only one of the pings. We've got 10.20.1.8 sending a ping to our 8.8.8.8 host out on the internet, and we can see here that the playbook's criteria. What is going to match our query in this example is when we ping 192.168.1.175, that will trigger the playbook and block the IP address. That's the difference here.

02:00-03:00
We configure this, but then we wait for the conditions to be met and the automated playbook will block that IP. Our test number four will use ATH, or automated threat hunting, playbook to block the host, 10.20.1.8, using the firewall rule on the Sophos firewall using the Sophos firewall connector. We start here by navigating this time to **Respond** and **Automation**. To create a new playbook, we select the **Create** button. That opens our window. There are four pieces, or four segments, or four sections to our playbook. The first one is the **Alert Configuration**, the second is the **Query Configuration**, the third, the **Condition Configuration**, and the final piece is the **Action**. It's the **Action** where we will choose to block the firewall, to block an IP address in the firewall.

03:00-04:00
The **Alert Configuration**: We're just going to go through the pieces here that are not default. In the **Alert Configuration**, yes, you have to give it a name. That is how we identify it in the table. The schedule, by default, is an interval. It is run every five minutes over all tenants. That is also a default. The schedule is enabled by default, and mute is disabled by default. We don't need to touch any of those in the configuration. Then for **Tenants**, we need to choose which tenants we are going to run this on. In our example, we're going to run it on all tenants and the indices that we're going to run this on is the **Traffic** indices. The reason that it is important to choose the right index is depending on what your query is that you're going to use to match and trigger this playbook. We're going to make sure that that query is also looking at data in the right index. In this example, our ping of 192.168.1.175 is a **Traffic** index.

04:00-05:00

We'll have an Interflow record come in alerting us of that ping that will trigger the, sorry, that will match the query associated with this playbook and trigger the actions. The index is **Traffic**. The **Rule Type** in this example is **Query**, and that is by default. Okay. The Query Configuration here's what we need to do here. There are a couple of different ways you can do this. My suggestion, before we go through this piece of the material, is I would create the query outside of the playbook and test it to make sure that it is matching the data you're expecting it to see. One of the difficulties of creating the query inside the playbook, is that you don't know if you've created it correctly. You don't know is the query matching a problem or is there simply no data matching that query? I always like to create the query outside of the playbook.

05:00-06:00
You can go to **System | Queries** and create it there. You can go to **Investigate | Threat Hunting** and create it there. Navigate to **Investigate | Threat Hunting**. Find the data you're looking to match, build the query, and test that it matches that data. After that's done, you can come in here to your ATH rule and simply bind the query that you've already created. But in this example, we're going to see how to create the query from within the playbook. We start by creating a new query. We give it a name, and then we start putting in our matching criteria. We can see we're going to match on an ICMP type is echo, and we're going to match for a destination IP is that 192.168.1.175. You can see in the query it is an "AND," so it must match both of those elements of our query. After you've done that, you must save the query. Don't forget to do that. Then the conditions.

06:00-07:00
This is a condition that we can use for quite complex ATH rules. We're not going to go into the details here. There's another lesson on ATH playbooks in our curriculum. Here, we're just going to keep it simple. We're going to give this condition the name of "default." It's not a special name. I could just as easily call that condition "David," and we'll see what that pops up down in the **Actions**. In the Action section, you can see here we've configured two actions: We've got an email action and a firewall blocking action. The email actions at the top you can see we're going to send an email to me that's going to include a copy of the Interflow record, and then the second action is a firewall action to block. We can see here using the Sophos firewall. After we choose **Block** from the radio buttons, then we choose from the drop down the Sophos firewall, then we choose **All Source IPs**.

07:00-08:00
You can see here we're not typing in any IP addresses like we did before. In the first two tests, we typed in the IP address. In the third one, we selected the IP address from the Interflow record. In this example, it's just saying, "I'm going to use the source IP in the record that has matched the query or or a destination IP or both," is your other option. We need to choose a direction. The **Direction** is **Incoming**, **Outgoing**, or **Both**. And a **Duration**, as we said we can be five minutes, five hours, five days, or forever. Okay. After we've saved that, sorry, we didn't see at the very end of that. When we get to the bottom, we click **Submit** to save that ATH playbook. We can see there it is on our list.

08:00-09:00
Our host, 10.20.1.8, will be able to communicate freely through the firewall as long as it doesn't ping, because that's that echo we were looking for, doesn't ping 192.168.1.175. As soon as it does that, the

Modular Sensor will send an Interflow record to the Data Processor, and the ATH playbook will be triggered. Our Modular Sensor is doing traffic monitoring. It has the **Traffic Monitor** feature turned on in the sensor profile for that Modular Sensor. It is watching the wire. It will see that ping go from our .8 host to the .175 host, and that will trigger our ATH playbook. Then the ATH playbook–here's the action: the ATH playbook will use the Sophos firewall connector to communicate to the firewall and add the source record, or the source IP of the record that matched the query to the source host IP group of that firewall rule.

09:00-10:00
Let's go in and have a look. If we go to **Respond | Actions** and select the **Firewall Actions** tab, we can see that our ATH playbook has been triggered. We can observe the initial add and the remove after five minutes of those records in that table. Note the **Status**. We'll go from waiting, to in progress. If you keep clicking that update button, it'll continually update the table. We can see it going from waiting, to in progress, to succeeded. After it succeeded, let's go have a look at our host to see what has happened. We know in the playbook we had two actions: We had the email, and we had the trigger to block. Here is our email that we've received. We've received it to my address and you can see we have our body of the email and we have attached the JSON record of the Interflow record that matched the query that triggered the ATH playbook.

10:00-10:50
That's the email resulting in it. Then if we wait, a period of time, the duration was five minutes, you can see that we're back up and running again on our host. After that time expires, the follow-up communication, so the connector removes the IP address, so it will send another API request to remove the source IP address in the Interflow record that matched the query that triggered the ATH playbook. All communication is good to go again.

The lesson, *Connector Verification - ATH Action*, is complete. Thank you.

L06B-DEMO: Connector Verification - ATH Action (21m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Connector - Sophos Firewall* module is *Demonstration of Connector Verification - ATH Action. Demonstration of Connector Verification. - ATH Action.* This demonstration shows you how to automate the process of using the connector to block the IP address of a host based on specific criteria being met. To do this we'll be using an ATH, or automated threat hunting, playbook.

Let's start with a look at the network diagram shown as the topology devices and host that we're going to use for this test. Alright. Test number four: What we're going to have is we're going to have our host 10.20.1.8. We're going to have it send in a ping to 8.8.8.8, but only that one. I know we've been sending two pings before, but only going to send it out to that internet host 8.8.8.8.

01:00-02:00

Then we're going to create an ATH playbook, an automated threat hunting playbook, that is going to use the criteria of a ping to our other host 192.168.1.175. If that criteria is met, the action we're going to use it to block the host and send an email. That will all happen automatically. We don't need to pay attention to that as an analyst. We can just let the system do that autonomously. First up, let's check our hosts. Alright. We said we're only going to do this left-hand side one, the 8.8.8.8. We're not going to use this. This is going to be the trigger, the 192.168.1.175, will be the trigger. Enter our UI. Enter admin and our password.

02:00-03:00
Here we are in our UI. Now the first thing we're going to do, just good common practice, we have to create a query. If we were to go into **Respond | Automation**, create a new automation, and create the query in here, now of course you can create a new query in here, but our suggestion is always to create it first, and then bind it to this ATH playbook. The reason we say that is we want to create it and test it that it's going to work. There's a couple of ways you can get there. We can do **System | Queries** and create it from in here. That'd be one way to do it, or we could go to **Investigate | Threat Hunting**. Let's look at the data that we're creating the query for. Let's change the index from **Alerts** to **Traffic**. Because the query we're going to create is 192.168.1.175 as a destination, and we're going to use echo as the ICMP type.

03:00-04:00
If it sees a ping, an echo, or an echo reply to that host, that destination host, it will trigger. Now we can see that there is traffic in here. Let's just actually make sure that that works. We could come in here and do a dstip of 192.168.1.175. Let's test that. Yes, we have data for that. We know that query's working so far, and (capital A-N-D) AND an ICMP_type of echo. That didn't work. It didn't work because I didn't spell "type" correctly: t-y-p-e. Alright. Then we can see that's in there.

04:00-05:00
If you didn't know what that field was, we could simply say the destination of 175, we know we could do "AND proto." We know that ICMP is protocol one. We could do that. Jump into one of these records, open it up, and look for reply. I'm going to do a browser find and sorry and look for an echo E-C-H-O. The reason that didn't show up there, that echo, what I'm looking at here is actually a firewall record where we're looking for a Modular Sensor record. We could be a little more clever and say, filter for Modular Sensor. If I open up this and look for echo, we see a whole bunch of them in here. Okay. The one I'm looking for is ICMP type. I could copy that as a query. Let's do that. Copy as query. Go back up to the top, get rid of our protocol one, paste our query.

05:00-06:00
We're only going to look for echo, so let's remove that. We can remove the quotes because we don't want the reply. We could just use a small "e." Then that's how we find our field ICMP_type echo. We know that there's one filter in here. Let's remove that filter for Modular Sensor, close that. Okay. I have some records here for our search. Okay. We know that that's correct. We know that's going to work. Now we're going to want to create a query for that. I could either copy that search field, come into **Filters**, open up **Queries**, and create a query. We create a new query. We give it a name, let's call this

“Ping 192.168.1.175.: I could come in here and just now just paste that string in that I copied from my query. That would be one way to do it and I know that would work.

06:00-07:00
We could say save. I could apply that query. If I come up here I can see there's my query ping. Let's remove it from up here and we can see we've now got a million records. I add my query to this filter, and we're back to our 543. I know my query is working. The other way we could have edited that query is let's go ahead and actually we didn't need to leave **Investigate | Threat Hunting**. **Investigate | Threat Hunting**. We were on the **Traffic** index and we had our query of ping. We could edit straight from there. If we edit this, that's what we pasted in there, the query string. If you wanted to do this differently, the other way we could do it is not to do a query string, but just to simply add a condition.

07:00-08:00
The condition we'll add is a field of DST IP is, we could say is exactly, 192.168.1.175, and save that. We've got some records. We can see that has worked so far, but let's edit and add the second piece. Select the **Add Condition** button. This one was icmp_type. There it is. String is echo. Save that. There we are at our 543 records we expected to see. Our query is bound. We know our query is working. If we remove the query from the filter, we get a bunch of records. We add our query to the filter and so we know it's working. Okay. Select **Clear all**. Now we can go and create our ATH playbook. **Respond | Automation**, and we can come in here. This is where we create the automations. We can see there's already four of them listed here.

08:00-09:00
We're going to select the **Create** button, and then give it a name. Let's call it “Block 10.20.1.8 on ping of 192.168.1.175.” Okay. That's the name of it. We're going to schedule this. The interval is going to run every five minutes over all our tenants. The tenants we're going to choose is **All tenants**. It is not the indices. The index is not **Alerts** but running this on the **Traffic** index. We choose **Traffic**. The rule type is a query. Now we've already created the query. We could create a new one here and go and add all those things we wrote here, but I always like to test that the queries work before you bind it. After it's created, it's just in the drop down. There it is. Our ping, the query ping. We can see the term there. We can edit it straight from within here again if we wanted to check it, but we know that's worked, so no need to change that.

09:00-10:00
Now the condition, this is just a name You can type any name in here. This name just gets bound between this condition and the action. Let's just put in default here. We can see now default just appeared down here. What this condition is saying is as long as we have at least one match to this query, then this condition name “default” will be used down here. Our action: So the first action we're going to do is our firewall action. We're going to choose our firewall, our Sophos firewall. Remember we're not typing in addresses here. This is saying whatever matches that query, what are we going to do? We're going to block all source IP addresses. This is the host. We want to block our host 10.20.1.8. When it pings that 192.168.1.175. We're going to block all source IP addresses.

10:00-11:00
You could choose to block both addresses, both source and destinations. We're going to block it in an incoming. This says that that host, 10.20.8, will not be allowed to communicate through the firewall, and we're going to do that duration of five minutes. Look, you could do it forever. There is a forever duration, which means that once that block goes in, you would have to manually undo it or have your firewall admins go and remove that address from that IP host group. We're not going to do that. We're going to do five minutes, and then send the block. That's one of our actions. Let's send another action. I want to receive an email. Recipients: I don't have a recipient in here, so I can just type in my email address. Okay. There's our email address that is spelled correctly.

11:00-12:00
The subject will be 10.20.1.8, pinged 192.168.1.175. That's what we want to say. We're going to give it a high priority because we really want to know. Our message is to our SOC analyst, dear admin or dear SOC analyst, the host pinged again. Alright. This **Include Interflow** record will attach a JSON to this record. Okay. Now we've done that. Our **Submit** button has shown up, so we're going to submit. We can see that it is active. Now I can wait. Remember we said it runs every five minutes. We don't quite know where in that five minute cycle it is. Let's just select **Run Now** to force it. But just before we do that we're going to go back to our host.

12:00-13:00
Our 8.8.8.8 is happily pinging along. We are not pinging the 175. Come back here and we're going to select the **Run Now** button. It's being triggered. It's going to check to see if there's any traffic that matches that. We're not yet pinging, so there shouldn't be any traffic that matches that. Our host is still happily pinging away to 8.8.8.8. It's a little hard to see. We can see that tracking up just there. Okay. Now what we're going to do is initiate this ping and this should then force the issue. We're going to send that ping there. We're going to come back. We're going to run it again to say go and check. What we could do just before we do that, let's go check let's go **Investigate | Threat Hunting** and see if there's any traffic matches. Let's close that filter, **Alerts**, turn that off, go to our **Traffic** index and let's apply our query.

13:00-14:00
Let's just look at the last 15 minutes. We can see there has been some traffic. Just in that last 15 minutes we have some traffic. We've got a destination of 150. Do we have the destination of the 192.168? We could add that up here. Our query is correct. Also, I've picked the wrong query. There is our query. Ping. Alright. There's some traffic. It's now come in. We can see we have an ICMP from that source to that destination. Our reputation's good, and we could. We don't need to see any more than that. We know that that's happened. Now that we've got that traffic let's go back to **Respond | Automation**, here's our threat hunting rule at the top. We're going to run it now rather than wait for that five-minute cycle. It's been run. We can see here it actually triggered.

14:00-15:00
What that is telling us is that there was something that matched, which is why it was triggered. If we go back to our host and have a look at what's going on, we can see there our host has just timed out. It's being blocked. Now the way we can now check what's going on in here is a **Respond | Actions |**

Firewall Actions, and we can see this has been added. We know that that's been added by our ATH playbook. Let's see what that looks like on our firewall. We know we blocked the host as a source. We should be able to go into our IP host group, that's Stellar.source IP host-group,

15:00-16:00
which is down here **Hosts and services**, **IP host group**, which should look at our source one and we should see a record in there: 10.20.1.8. That's how that arrived in there. Let's go back to our UI. We're going to revert this. We're going to undo this. We won't wait for the five minutes to time out. Click **Revert**. That is now being sent to the firewall. We're waiting. It's now in progress. This has been communicated to the Sophos firewall. The final one is it has succeeded. Alright. Let's go look at our host. Our host is happy. It's pinging again. That's great news. If we go to our firewall, and look in that group again, the source, it is now empty. That's all good.

16:00-17:00
The final thing we could probably do is just go look at our logs in here, let's not look at our firewall logs. We want to look at our administrative logs, and we should see a bunch. We can see here. Look at that! IP hosts were deleted by admin from 10.20.1.15. That's because we're running our connector on our Modular Sensors. That's why that's coming through there. This would be us logging in here and these will be the other ones coming. You can see this change. We don't see our actual host IP address put into this record. Even if you look at the **detail** it doesn't show you that. We don't look at all of them. Sorry, when we check **detail** it then goes and adds all of them, but we can see that they were deleted. We do see it here. We see our IP host group name. That'd be part of the API call. The one we're sending it to is the Stellar source dash group. Sorry that that goes over a line.

17:00-18:00
Alright. Is everybody happy again? Are our hosts pinging? Now this will run again, right? If we go back to **Respond | Automation**, this is running. This is still enabled. We can see that this was triggered at 9:50:05. I don't know if we've been doing things long enough to wait for that five minutes to roll around, but let's just see because if we look at the host, the host is still pinging 175. The next time this is triggered, and we know that that runs every five minutes, then our host should get blocked. Let's just wait a few minutes and see if this triggers again. I'm going to just refresh this table to see if the last triggered goes from 9:50 to a later time. Maybe it's something before 9:50:05. And there we go!

18:00-19:00
It was triggered again at 9:53. We go to our host and we can see our host has now been blocked. All of those things will be back in line. **Respond | Actions | Firewall Actions**. There it is. There it is sitting there. It's another incoming block for five minutes. If we go back to our firewall, we will see, we'll just close our log view here, and we'll update this so I'm going to cancel and then look again. There it is again. As long as this host is pinging 192.168.1.175, it is going to trigger in our ATH rule. If we go and edit that it's going to trigger because it matches this query. We're looking every five minutes in the **Traffic** index to match that query.

19:00-20:00

If it does, it's going to send an API call to our connector, the Sophos firewall, and it's going to send us an email. Let me also just check for some emails. One moment. I think my email is closed. One moment. There is our email that we received at 9:55. That was the second one. Here's the first one. This was at 9:50. We can see here as we expected that's the subject of our email that we typed in. It was sent to me and reads, “Dear SOC Analyst, The host pinged again…” and we get an attachment of our Interflow records. Let's open that up. Here is our Interflow record. This is the record that triggered or matched the query that triggered the ATH playbook. The engine ID name is our Modular Sensor.

20:00-21:05
That is what picked up this with an app family name of network, so that was our network traffic. This is the Modular Sensor’s feature looking at the traffic on the network. There is our ICMP type echo reply, so that's the match. Here’s our destination IP address of 192.168.1.175. There is our match for our query. Before we go, there is one more. It is now 10:02, but this is the 10 o'clock. We can see this every five minutes, this is going to trigger. That's how you configure an ATH rule to block an IP address using a connector.

The lesson, *Demonstration of Connector Certification - ATH Action*, is complete. Thank you.